

Applied AI Research Engineer

Summary

Applied AI Research Engineer in computer vision and model efficiency. First-author arXiv researcher and merged Hugging Face PEFT contributor. Designs controlled multi-seed experiments and carries research from hypothesis through evaluation and deployment benchmarking.

Research

- 2026 **Student Capacity Moderates Knowledge Distillation Effectiveness**, *First Author* — [arXiv:2605.31191](https://arxiv.org/abs/2605.31191)
Logit-KD vs Feature-KD across four ResNet teacher–student pairs, leakage-free model selection, five seeds. Statistically significant KD gains occurred for higher-capacity students, while larger teachers at matched accuracy did not increase gains; Feature-KD matched or exceeded Logit-KD across all pairs. A controlled v2 re-run rejected an earlier bug attribution, tracing the discrepancy to test-set model selection.
arxiv.org/abs/2605.31191 github.com/umutonuryasar/kd-capacity-gap

Selected Projects

- 2026 **MicroCLIP — From-Scratch Vision–Language Model**, *Independent Research Project*
CLIP-style dual encoder built from scratch (ResNet-18, transformer text encoder, 16K BPE tokenizer). Controlled study of softmax InfoNCE vs SigLIP-style sigmoid on COCO Captions (batch 128 and 512 with three seeds each, plus single-seed 256): softmax matched or outperformed sigmoid at every batch size and, contrary to the initial hypothesis, the gap widened as batch size shrank. Shipped as an in-browser int8 ONNX demo; all 20 trained checkpoints are public.
PyTorch, Vision–Language Models, Contrastive Learning, COCO, Transformers github.com/umutonuryasar/microclip
- 2026 **RT-DETR Knowledge Distillation — COCO**, *Independent Research Project* — *v1.0 complete*
End-to-end framework for distilling a lightweight RT-DETR: five KD method families (eight loss variants) across 11 controlled configurations on a 30K-image COCO subset with leakage-free selection and multi-seed evaluation. Best configuration lifted student mAP by +74% (0.0388 to 0.0676) in this regime; follow-up controls showed loss weighting, not the proposed curriculum, explained the apparent gain. fp16 on RTX 3050: 160.8 FPS, 64 MB VRAM, no mAP loss.
PyTorch, Object Detection, Transformers, COCO, CUDA github.com/umutonuryasar/rt-detr-kd

Open-Source Engineering

- 2026 **Hugging Face PEFT**, *Merged PR 3293*
Fixed CUDA allocator overhead in `prepare_model_for_kbit_training`: temporary fp16-to-fp32 casts left approximately 1 GB of reserved-but-unused CUDA memory on constrained devices. github.com/huggingface/peft/pull/3293
- 2026 **aisuite**, *Merged PR 319*
Restored Python 3.14 compatibility by relaxing the `docstring-parser` constraint that pulled in removed AST APIs.
github.com/andrewyng/aisuite/pull/319

Artifacts huggingface.co/umutonuryasar

Experience

- 2025–Present **Applied AI Research Engineer**, *Independent Researcher*, Ankara, Türkiye
- 2024–2025 **AI Trainer — Freelance**, *Outlier*, Remote
Evaluated and ranked LLM outputs for RLHF-oriented data pipelines; authored and reviewed prompts, reference responses, and technical evaluations across programming, mathematics, and engineering.
- 2020–2024 **Freelance Software Engineer**, *Self-Employed*, Ankara, Türkiye
Project-based signal processing and RF software for clients, from algorithm prototyping to C++/Qt desktop tools.
- 2014–2015 **Engineering Internships**, *HAVELSAN (2015); TAI — Turkish Aerospace, Avionics (2014)*, Ankara, Türkiye

Education

- 2020 **B.Sc. Electrical & Electronics Engineering**, *Bolu Abant İzzet Baysal University*, Bolu, Türkiye

Technical Skills

- ML Computer vision, vision–language models, transformers, contrastive learning, knowledge distillation
- Efficiency PEFT and LoRA, int8 quantization, mixed precision, memory optimization, inference benchmarking
- Tools PyTorch, Hugging Face Transformers and PEFT, OpenCV, NumPy, CUDA, ONNX, FastAPI, Docker, Linux
- Programming Python, C/C++ (Qt), Bash, MATLAB, Git, CI and testing, digital signal processing
- Languages Turkish (native), English (full professional proficiency)